SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

MTH210

Examination - July Semester 2025

Fundamentals of Probability

Thursday, 13 November 2025 04:00 pm - 06:00 pm

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FOUR (4)** questions and comprises **FOUR (4)** printed pages (including cover page).

2. You must answer **ALL** questions.

3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.

4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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Answer ALL questions. (Full marks 100)

Question 1

(a) The probability that a company will hold a sales promotion for product A is 0.7. The probability that it will hold a sales promotion for product B is 0.4. The probability that it will hold a sales promotion for either product A or product B or both is 0.8. Calculate the probability that the company will hold a sales promotion for neither product.

(4 marks)

(b) Table Q1 below shows the data of 100 individuals collected from a survey where respondents were classified based on their smoking status and whether they had lung disease.

	Smoker	**Non-smoker**
Lung disease	30	5
No lung disease	10	55

Table Q1

If an individual is randomly selected, solve the probability that this person has lung disease, given that the person is a smoker.

(6 marks)

(c) The probability that a student passes Maths (A) is 0.7. The probability to pass Physics (B) is 0.6 and the probability to pass Chemistry (C) is 0.8. It is also given that the probability that a student passes both Maths and Physics is 0.5, and the probability to pass at least one of the three subjects is 0.90. Solve $P(A|A \cup B \cup C)$.

(6 marks)

(d) Of all the women who submit pregnancy tests, 75% of them are pregnant. A certain pregnancy test kit has a probability of 0.02 of giving a false positive result (i.e. predict that a woman is pregnant when in fact she is not pregnant). The test kit has a probability 0.99 of giving a valid positive result (i.e. predict positive when the woman is really pregnant). Compute the probability that a woman is not pregnant given that the test is negative (i.e. test predicts that she is not pregnant)?

(9 marks)

Question 2

(a) There are six telephone lines at a company reception counter. Let X be the number of lines in used at a certain time of the day. The probability mass function (pmf) of X is given in Table Q2 below.

x	0	1	2	3	4	5	6
$P(X = x)$	0.10	0.20	0.15	s	0.20	0.06	0.04

Table Q2

(i) Compute the probability that between one and four lines, inclusive, are in use.

(6 marks)

(ii) Determine the standard deviation of X.

(7 marks)

(b) Suppose Andrea rolls two dice, and Andy pays Andrea the dollar amount of the difference (X) in the scores. Solve the probability distribution of X. Analyse and compute the minimum amount that Andy should charge Andrea for a game so that he can make a profit in the long run (i.e. playing many rounds of the game)? Round off your answer to the nearest dollar. (Hint: Consider the probability distribution of X and its expected value.)

(12 marks)

Question 3

(a) Mr and Mrs Loo decide that they will continue to have children until they have two females. Assume that the probability of having a female child is 0.5. The couple would like to find the probability that their second female child is their seventh child. Mr Loo said that a geometric model can be used to find the desired probability, while Mrs Loo argued that using a negative binomial model is more suitable. Discuss who is correct. State two conditions required for the model to be appropriate for application. Applying an appropriate model, compute the probability that their second female child is their seventh child.

(10 marks)

(b) The probability that an applicant will pass a pre-requisite assessment for a course in each attempt is 0.7.

(i) Determine the probability that an applicant will pass the assessment in at most 3 attempts.

(5 marks)

(ii) In a certain round of the course application, there are 8 applicants taking the pre-requisite assessment. Analyse the problem and determine what probability model can be applied to solve for the probability that at least four applicants will pass the assessment in the first attempt.

(10 marks)

Question 4

(a) Determine the value of the constants b and c in the probabilities

$$P(Z \leq b) = 0.121 \text{ and } P(-c \leq Z \leq c) = 0.668$$

where $Z \sim N(0, 1)$.

(10 marks)

(b) In a supermarket, the mass in gram of a packet of sweets has the distribution $X \sim N(120, 15^2)$, and the mass of a packet of chocolate has the distribution $Y \sim N(180, 20^2)$. X and Y are independent. Calculate the probability that a randomly selected packet of sweets weighed less than 130 grams. In addition, compute the probability that a randomly selected packet of sweets and a randomly selected packet of chocolate have a total mass of at least 310 grams.

(15 marks)

----- END OF PAPER -----